Hierarchical Feature Fusion Triple Network for Change Detection With Bitemporal Remote Sensing Images

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***Abstract*— Achieving land cover change detection (LCCD) through remotely sensed images (RSIs) is important in the observation of the changes on the Earth's surface. In such detection, spectral-reflectance noise and the uncertainty of the imaging external conditions for the bitemporal RSIs usually cause some salt-and-pepper noisy pixels in the results and reduce the change detection accuracy. In this article, a hierarchical feature-fusion triple network (HFTN) is proposed to improve the performance of LCCD with RSIs. Overall, the proposed HFTN aims to learn representative features to improve change detection performance via two feature learning enhancement strategies and a hierarchical feature-fusion mechanism. First, an image feature difference model is proposed to generate the input feature for the middle branch and guide the learning performance. Second, a progressive denoising module (PDM) is proposed and applied to each temporal image to reduce the noise before feeding the features into the backbone of the proposed HFTN. Finally, a hierarchical feature-fusion module (HFFM) is proposed to fuse the learned deep feature for generating a change-magnitude image. Additionally, multiscale convolution, cross-scale fusion, and a shared weight are adopted in the backbone of the proposed HFTN to further enhance the feature learning performance. Compared with eight state-of-the-art methods, experimental results verified the feasibility and superiority of the proposed HFTN for LCCD with RSIs. For example, the proposed HFTN achieved improvement rates of approximately 0.43%–11.83% for overall accuracy (OA) and 0.11%–4.81% for false alarms (FAs) across six pairs of real RSIs. The code can be available at https://github.com/ImgSciGroup/HFTN-NET.git.**

***Index Terms*— Land cover change detection (LCCD), neural network, remotely sensed images (RSIs).**

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I. Introduction

LAND cover change detection (LCCD) with remotely sensed images (RSIs) is an important technology for observing Earth's surface changes [1], [2], [3]. Owing to the advantages of remotely sensed technology in effectively capturing large-scale land cover [4], [5], [6], LCCD with RSIs has been widely used in land-use motoring [7], geological disaster detection [8], [9], natural resource management [10], and environmental monitoring [11]. However, LCCD usually refers to more than two or three temporal RSIs in a detection task, and some challenges are usually encountered in practical applications, as presented below.

1) Spectral noise is inevitable in an imaging process for RSIs [12]. The imaging progress depends on two aspects: the platform and the sensor. Intuitively, preprocessing operations, such as geometry and reflectance correction, are required for a raw RSI acquired with a remotely sensed platform (such as a satellite or an aerial plane). This operation may smoothen some noise or bring some noise to a processed image [13], [14]. Additionally, remote sensors belong to optical physical equipment, which produces some noise in the imaging process. Moreover, the noise from the equipment is inevitable.
2) Pseudo-change is observed as noise in a change detection map when using bitemporal RSIs [15]. The bitemporal RSIs used for LCCD cover the same geography but occur on different dates. External imaging conditions, such as atmospheric moisture, phenology difference, and sun height, cause pseudo-changes in an LCCD task [16], [17]. Additionally, mixing pixels may also occur in RSIs and result in accused pseudo-changes [18]. These pseudo-changes may be considered as noise in a detection map and reduce the detection accuracies.
3) Ground targets on the Earth's surface with various shapes and sizes challenge the utilization of spatial-contextual features. Many studies have demonstrated

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that utilizing spatial-contextual features can improve the application performance with RSIs [19]. The ground targets in an image scene with different shapes and sizes result in changed and unchanged areas between bitemporal RSIs, also with varied shapes and sizes. Utilizing spatial-contextual features for LCCD with RSIs is a challenge.

To deal with these challenges for LCCD with RSIs, various methods have been proposed in the past decades [2], [20], [21]. Given the advantages of deep learning (DL) techniques to explore deep abstract features, DL has been applied successfully in computer vision [22], signal processing [23], object detection [24], and image segmentation [25]. This article focuses on the DL algorithm in LCCD with RSIs.

The most popular DL techniques is the basic backbone called convolutional neural network (CNN), which has been widely adopted in the field of LCCD with RSIs [26]. CNN concentrates on exploring deep abstract features by applying a series of convolution operations on feeding datasets. Many approaches based on CNN, such as 2-D CNN [27], 3-D CNN [28], axial cross-attention branch fusion networks [29], and feature difference CNN [30], have been promoted and applied successfully in the community of LCCD with RSIs. In addition, because the LCCD task refers to bitemporal RSIs, Siamese neural networks have been constructed to improve the LCCD performance [31], such as semi-supervised Siamese network [32], Siamese nested UNet (SNUNET) [33], Siamese convolutional network [34], and fully convolutional Siamese network [35]. Many studies indicate that CNN can explore the deep features from bitemporal RSIs and smoothen the noise of change detection, but the convolutional operation with a regular shape and a single kernel cannot cover many changed and unchanged areas with various shapes and sizes. To consider many ground targets with various shapes and sizes, multiscale convolution was an intuitive way and applied widely in LCCD with RSIs [36]. In this approach, multikernels are coupled with CNN to explore multiscale deep features for change detections, such as multiscale CNN [37], pyramid deep feature learning networks [38], [39], multiscale U-Net [40], and multiscale fully convolutional Siamese network (MFCN) [41]. The reviewed studies with multiscale CNN show that extending convolution with multiscale kernels can cover many ground targets and improve the change detection performance.

In addition, a transformer neural network is also attractive in LCCD with RSIs for use in sequence data processing and global information capture [42]. Transformers depend on the parallel multihead attention mechanism, which is notable for requiring less training time than many existing recurrent neural networks [43]. Given the advantages of the transformer neural network, many approaches based on transformers have been proposed for LCCD with RSIs. For example, to capture long-range feature dependencies, the multiscale transformer [44], [45] and the attention-based transformer [46] have been proposed to use the information captured by the network structure for feature information enhancement. Moreover, to utilize the contextual information of the transformer, a fusion-based transformer network [42], [47], [48] has been developed to improve the accuracies of LCCD with RSIs. Furthermore, LCCD with RSIs usually involves bitemporal images, so the Siamese network and the transformer are combined to improve the coding ability of high-order semantic features [49], [50].

In recent years, feature fusion has become increasingly popular in application of the DL neural network for LCCD with RSIs [29]. Feature fusion concentrates on fusing the learning features from different temporal images to generate a change magnitude image and then obtain a binary change detection map [51]. For example, a supervised image fusion network has been widely adopted for LCCD [52], [53]. In the process of supervised image fusion, the training progress needs a ground reference map to guide the feature-fusion direction and outcome of the targets in a fused detection map. Luo et al. [54] also introduced a unique diff-feature contrast thought that innovatively combines a small number of labeled samples and a large number of unlabeled samples to train the network simultaneously, improving the generalization performance of the network. However, basic CNN, which is a supervised image fusion network with a single-size convolutional kernel, cannot cover many ground targets with various shapes and sizes. To cover more ground targets with various shapes and sizes, multiscale feature-fusion networks [55], [56], [57], [58] and adaptive feature-fusion networks [59], [60] were promoted to improve the performance for LCCD with RSIs. In addition, to enhance the feature-fusion performance, multifeature fusion networks [61], which concentrate on fusing multiview or multisourcing features to improve the distinguishability of ground targets, are promoted for LCCD with RSIs. Various studies indicate that combining feature fusion with DL networks can enhance the distinguishability of ground targets and improve the change detection performance.

Inspired by the existing studies, a hierarchical feature-fusion triple network (HFTN) to enhance the change detection accuracies and performance is proposed in this article. The contributions of this study are summarized as follows.

1) A novel approach named HFTN is proposed in order to enhance the performance and accuracies of LCCD with RSIs. Compared with many neural networks, the HFTN contributes the idea of fusing the learned information to supplement the lost information in the training process, and a hierarchical feature-fusion module (HFFM) is proposed to conduct the idea.
2) A simple yet effective progressive denoising module (PDM) is first proposed to smoothen the noise from the input image feature. Intuitively, remotely sensed imaging often has various types of noise, which usually have a negative effect on the learning performance of a neural network. Here, a PDM is proposed to reduce the negative effect and improve the change detection performance. Although the PDM has been verified in LCCD with RSIs, it may be preferred for other applications with images.
3) Advanced change detection performance can be achieved by the proposed HFTN. Compared with eight

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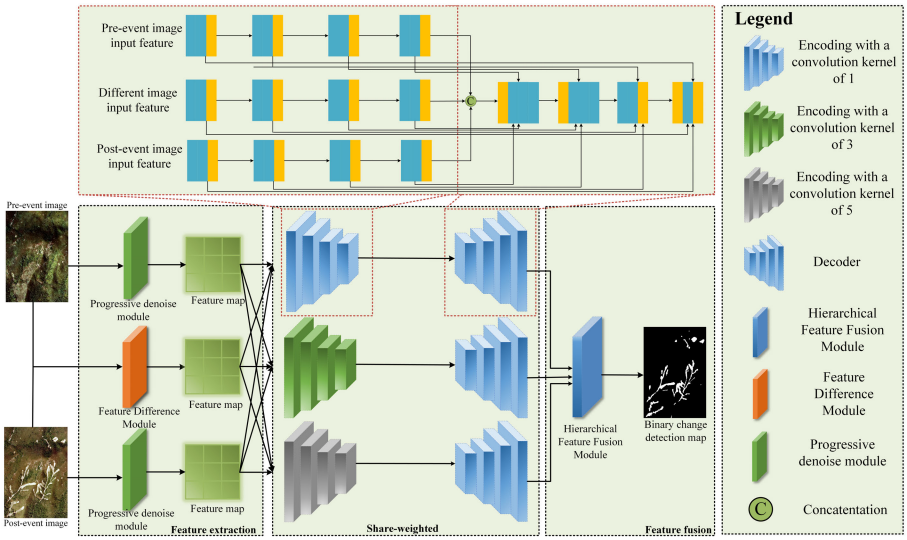


Fig. 1. Flowchart of the proposed HFTN.

state-of-the-art methods on six real RSIs, the experimental results indicate that the proposed HFTN achieves superior change detection maps for visual performance and quantitative accuracies. In addition, the ablation experiment further demonstrated the effectiveness of the proposed HFTN's modules.

The rest of this article is organized as follows. Details of the proposed HFTN are given in Section II. The experiments are introduced and presented in Section III. Finally, the conclusion is summarized in Section IV.

II. Methodology

In this article, an end-to-end network named hierarchical feature-fusion triple network (HTFN) is proposed for LCCD with RSIs. First, an overview of the proposed HFTN was given to provide an entire impression of the flowchart of the proposed approach. Then the three proposed submodules named feature difference module (FDM), PDM, and HFFM are introduced in detail. Finally, the loss function of the proposed HFTN was given to understand the training of the proposed network. Details of each part are presented below.

A. Overview of the Proposed HFTN

Fig. 1 shows the structure of the proposed HFTN. The structure is similar to the encoding and decoding of U-Net [62], but the proposed HFTN is a triple network, and the encoder of the proposed HFTN is using shared weights aiming at reducing the number of parameters. Three modules, namely the feature difference, progressive denoising, and feature-fusion modules, are proposed and embedded in the backbone of the proposed HFTN to improve the performance of LCCD with RSIs. The algorithm flow of the proposed method is shown in Algorithm 1.

First, the bitemporal images are taken as the input for the PDM, which involves convolution and pooling. Then, the FDM obtains the difference image of the bitemporal images.

Algorithm 1 HFTN Algorithm Flow

Input: A set of image pairs $\{X_n^{t1}, X_n^{t2}\}$ from the N-th batch and the corresponding ground truth y_n, learning rate parameter η, training epochs N, Hyperparameter θ.

Output: Binary change detection map M between the two images.

Step-1: For epoch [1,N], do:

Step-2: let F_{diff} be the difference feature between X_n^{t1} and X_n^{t2}, $F_{\text{den1}} \leftarrow FDM(X_n^{t1})$, $F_{\text{den2}} \leftarrow FDM(X_n^{t2})$

Step-3: $F_{\text{diff}}^1, F_{\text{den1}}^1, F_{\text{den2}}^1 \leftarrow Conv((F_{\text{diff}}, F_{\text{den1}}, F_{\text{den2}}), K_1)$
$F_{\text{diff}}^3, F_{\text{den1}}^3, F_{\text{den2}}^3 \leftarrow Conv((F_{\text{diff}}, F_{\text{den1}}, F_{\text{den2}}), K_3)$
$F_{\text{diff}}^5, F_{\text{den1}}^5, F_{\text{den2}}^5 \leftarrow Conv((F_{\text{diff}}, F_{\text{den1}}, F_{\text{den2}}), K_5)$

Step-4: $F^1 \leftarrow Decoder(Encoder(F_{\text{diff}}^1, F_{\text{den1}}^1, F_{\text{den2}}^1))$
$F^3 \leftarrow Decoder(Encoder(F_{\text{diff}}^3, F_{\text{den1}}^3, F_{\text{den2}}^2))$
$F^5 \leftarrow Decoder(Encoder(F_{\text{diff}}^5, F_{\text{den1}}^5, F_{\text{den2}}^5))$

Step-5: $\hat{y} \leftarrow HFFM(F^1, F^3, F^5)$

Step-6: $L \leftarrow L_{\text{loss}}(\hat{y}, y_n)$
$\theta \leftarrow \theta - \eta \frac{\partial \mathcal{L}}{\partial \theta}$

Step-7: Return M

End

Second, the obtained denoised feature and difference maps are sent to the encoding stage, which actually includes three such stages, each containing three convolution kernels (1×1, 3×3, and 5×5). Each layer in the encoding includes two convolution operations and one max-pooling operation. Next, the feature information was explored by convolutions with different convolution kernels, and the encoding stage is upsampled. Finally, in the fusion stage, the feature information obtained by different convolution kernels in the decoding stage is fused to generate a binary change detection map. The details of each main process are detailed below.

B. Feature Difference Module

Change magnitude images are pivotal in achieving a change detection task, because they show the change magnitude for

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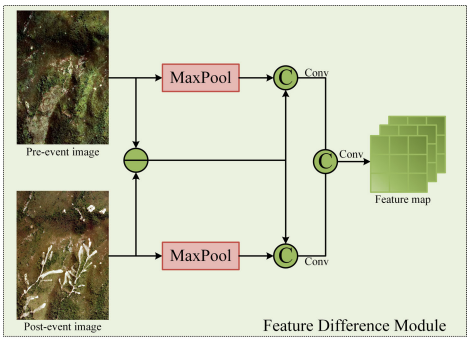


Fig. 2. Structure of the FDM.

each pixel in the image scene. In this section, an FDM is proposed to generate change-magnitude images and guide the learning progress of the proposed neural network. The structure of the proposed FDM is shown in Fig. 2. First, the max-pooling operation smoothens the noise of the bitemporal images X_n^{t1} and X_n^{t1}, and the difference operation is performed on those images to obtain a difference image F_{diff}. The equation is as follows:

$$F_{\text{diff}} = \left|\text{pool}(X_n^{t1}) - \text{pool}(X_n^{t1})\right|. \tag{1}$$

Second, the pre- and post-event images processed by max pooling are concatenated with the difference feature separately to enhance the spectral similarity, and the resulting feature information is processed using 3×3 convolution to obtain two features. Finally, the two features are obtained to be concatenated and processed by convolution with a 3×3 kernel to obtain a difference image. The proposed FDM can generate a change magnitude image F', which will be fed into the middle branch of the proposed HFTN to guide the learning performance. This process can be given as

$$F_1 = \text{Conv}\left(\text{concat}\left(\text{pool}(X_n^{t1}), F_{\text{diff}}\right), K_3\right) \tag{2}$$

$$F_2 = \text{Conv}\left(\text{concat}\left(\text{pool}(X_n^{t2}), F_{\text{diff}}\right), K_3\right) \tag{3}$$

$$F' = \text{Conv}(\text{concat}(F_1, F_2), K_3). \tag{4}$$

C. Progressive Denoise Module

Spectral noise in RSI is an inevitable phenomenon [63]. In the applications of LCCD with RSIs, spectral noise may cause pseudo-changes and noise in a change detection map. As shown in Fig. 3, in the proposed PDM, three convolutions with a kernel size of are applied to denoise the input image. After obtaining the feature map, three different pooling operations, namely AdaptiveavgPooling, AdaptivemaxPooling, and Maxpooling, are used for denoising and obtaining three denoised feature maps F_{a-avg}, F_{a-max}, and F_{max}. Second, three convolutions with a kernel size of 3×3 are applied to the denoised feature map, and the denoised feature map from different pooling operations is used to supplement the lost information in the convolutional progress. Third, three 3×3 convolutions in parallel are concatenated into two 3×3 convolutions in parallel, and two parallel 3×3 convolutions are concatenated into three 1×1 convolutions, and then the denoised feature map is generated. The equation is as follows:

$$F_{\text{den}} = C_{1\times1}(C_{3\times3}(\text{concat}(C_{3\times3}(F_{\text{a-avg}}), C_{3\times3}(F_{\text{a-max}}) \\ C_{3\times3}(F_{\text{max}})))). \tag{5}$$

The structure shows that the PDM achieves a progressive denoising performance, indicating gradual and continuous denoising, which helps smoothen the noise and preserve the invariant feature of bitemporal RSIs in the training progress of a neural network.

D. Proposed HFFM

In addition to the proposed PDM, an HFFM is proposed suggested and embedded in the backbone of the proposed neural network. As shown in Fig. 4, the HFFM consists of the convolution, batch normalization, and Relu functions [64], [65], [66]. First, the three feature information obtained by the backbone network is denoted as f_1, f_2, f_3, which are used as input to the HFFM. f_1 and f_2, f_1 and f_3, and f_2 and f_3 are fused. The three feature information is obtained by 3×3 convolution, batch normalization, and Relu function processing, and the three feature information is denoted as f_1', f_2', and f_3'. This process can be given as

$$f_1', f_2', f_3' = \text{ReLU}(\text{BN}(\text{Conv}_{3\times3}(f_1, f_2, f_3))). \tag{6}$$

Second, f_1' and f_2' are fused, f_2' and f_3' are fused, and the obtained fused feature information is processed by 3×3 convolution, batch normalization, and the Relu function. Finally, the two information types obtained are fused and processed by convolution, batch normalization, and the Relu function to obtain the feature information fusion map. Unlike many existing feature-fusion networks, the proposed feature-fusion module accomplishes feature tasks hierarchically, which is helpful in the processing.

E. Loss Function

In the proposed HFTN, a hybrid loss function, which includes mean-squared error and cross-entropy, is adopted. Here, hyperparameter λ_1 is adopted for balancing the affective mean-square error and cross-entropy. The formula of the loss function is as follows:

$$L = \lambda_1 L_{mse} + (1 - \lambda_1) L_{CE} \tag{7}$$

$$L_{mse} = \frac{1}{N}\sum_{i=1}^{i=N}(y_i - x_i)^2 \tag{8}$$

$$L_{CE} = -\frac{1}{N}\sum_{i=1}^{i=N}[(x_i \log(\sigma(y_i))) + (1 - x_i)\log(1 - \sigma(y_i))] \tag{9}$$

where λ_1 is the hyperparameter, and L_{mse} is the mean-square error as presented in (2); and L_{CE} stands for the cross-entropy loss function as shown in (3). Here, N represents the total number of samples, y_i is a prediction label for an unchanged or changed pixel, and x_i represents the reference truth value, which is either 1 or 0. Furthermore, $\sigma(y_i) = 1/(1+\exp(-y_i))$ is the Sigmoid function used to turn the output of a neural network into a probability value between 0 and 1.

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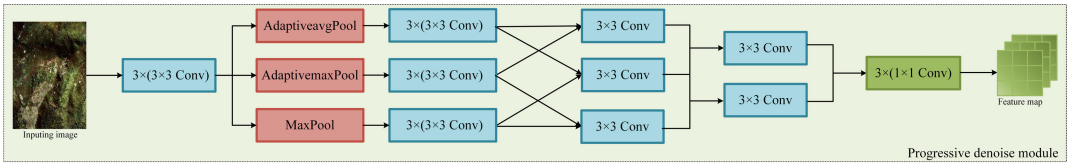


Fig. 3. Structure of the PDM.

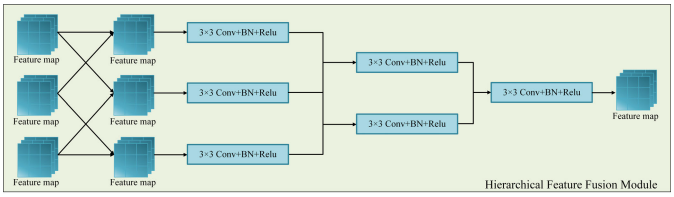


Fig. 4. Structure of HFFM.

TABLE I
DETAIL DESCRIPTION FOR EACH DATASET

Dataset	Year	Location	Sensors	Resolution
A	2007 and 2014	Hong Kong, China	Aerial camera	0.5 m/pixel
B	2006 and 2011	Hong Kong, China	Aerial camera	0.5 m/pixel
C	2004 and 2012	Jinan, Shandong	Quickbird	0.65 m/pixel
D	2004 and 2011	Jinan, Shandong	UAV	0.65 m/pixel
E	2001 and 2017	Hei Longjiang, China	Landsat	30 m/pixel
F	1995 and 1996	Sardinia	Landsat	30 m/pixel

III. EXPERIMENT

Three experiments were conducted to verify the feasibility and superiority of the proposed HFTN for LCCD with RSIs. The first experiment focused on comparing the proposed HFTN with eight state-of-the-art methods, including SNUNET [33], DSIFN [52], BIT_CD [48], MFCN [41], PA-Former [47], TFI-GR [67], DSAMNet [68], and USSFC-Net [69], aiming at demonstrating the performance of the proposed HFTN. The second experiment is an ablation experiment, which aims to verify the effectiveness of the modules in the proposed HFTN. The third experiment focusses on the parameters of the proposed HFTN to expand the applications of the proposed approach. The details of the experiment are described below.

A. Data Description

As shown in Table I, six pairs of real bitemporal RSIs were adopted in the experiments, and the details of each dataset are described below.

1) *Datasets A and B:* These datasets describe landslide events with aerial images (with a spatial resolution of 0.5 m/pixel) in Lantau Island, Hong Kong, China. Pairwise images were acquired in April 2007 and July 2014, 2006, and 2011 for datasets A and B, respectively. Both datasets A and B were acquired using an aerial camera.
2) *Datasets C and D:* These datasets describe land use change events in Jinan, Shandong Province, China. Dataset C was acquired by the Quickbird satellite with a 0.65 m/pixel resolution. Dataset D was acquired by an UAV with a resolution of 0.8 m/pixel and describes the land use change event near a railway.
3) *Datasets E and F:* These datasets were acquired with a Landsat satellite with a 30 m/pixel resolution. Dataset E describes the forest deformation in June 2001 and June 2017 caused by the wildfires, which occurred at the junction between Inner Mongolia and Heilongjiang Province, China. Dataset F describes the changes in Sardinia Lake in 1995 and 1996. Both datasets E and F were acquired using the Landsat satellite.

As shown in Fig. 5, the data refer to different change events, data types, and resolutions. Moreover, some bitemporal RSIs perform with strong phenomenon and illumination differences. Therefore, achieving a satisfactory change detection performance on the above datasets is a challenge, and these datasets are suitable for testing the robustness and performance of the proposed approach.

B. Experimental Settings

The experiments were conducted on a server with an NVIDIA GTX 2080Ti GPU, 12 GB of VRAM, and running the Windows 10 operating system. The proposed method is compared with eight state-of-the-art methods on six real RSIs to demonstrate the effectiveness and feasibility of the proposed method. The characteristics of each method are described in detail as follows.

1) *SNU-Net [33]:* This method has an embedded attention module in the classical U-Net++ that enables it to obtain enhanced detection performance. A compact skip connection is used in SUN-Net to supplement the feature information that is lost during the coding-decoding process. SNU-Net has an advantage in exploring the representative features from the different semantic levels and obtaining satisfactory binary detection maps with homogeneous RSIs.
2) *DSIFN [52]:* This approach is based on attention to extract multiscale deep features and uses differential features to supplement the information to train the network.
3) *BIT_CD [48]:* This method is based on the transformer network, which aims at capturing more context information via transformer encoder–decoder structure in the compact space-time on the basis of the token operation.
4) *MFCN [41]:* This method adopts a multiscale module and feature fusion in feature learning extraction.

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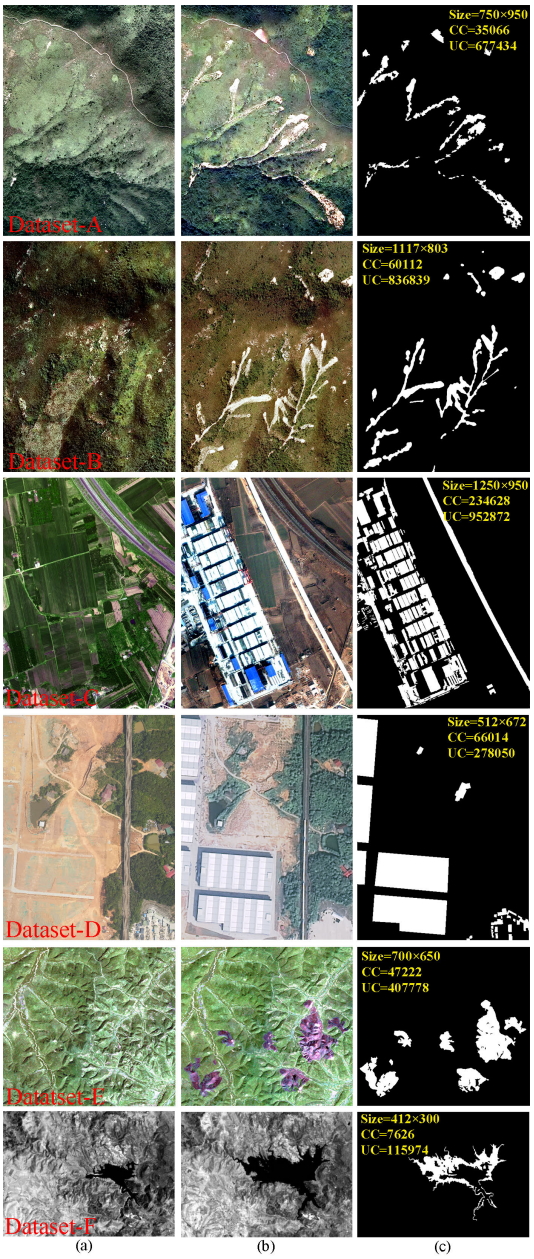


Fig. 5. Visual information for the six datasets (A–F) used in experiment 1. (a) Pre-event image. (b) Post-event image. (c) Ground reference map.

Moreover, the MFCN enriches the feature information through a skip connection.

5) *PA-Former [47]:* This approach extracts prior features on the basis of the transformer network to explore contextual tokens with spatial–temporal features from the prior information and integrates them into the deep features.
6) *TFI-GR [67]:* This network aims at capturing temporal difference information through temporal feature interactions and aggregates different low- and high-level temporal representations through the proposed guided refinement module to measure the change map.
7) *DSAMNet [68]:* The backbone of DSAMNet is the classical ResNet network, and it is coupled with deep supervision and attention mechanism and measures the change magnitude between bitemporal RSIs through a metric module.
8) *USSFC-Net [69]:* This method uses multiscale decoupled convolution and spatial–spectral feature cooperation without additional parameters to learn the change magnitude between bitemporal RSIs.

TABLE II
SUMMARY OF THE APPLIED EVALUATION INDICATORS

Overall Accuracy (OA)	$OA = \frac{TP+TN}{TP+TN+FP+FN}$
Average Accuracy (AA)	$\mathrm{AA} = (\frac{TP}{TP+FP} + \frac{TN}{TN+FN})/2$
Kappa Coefficient (Ka)	$P_0 = \frac{TP+TN}{TP+TN+FP+FN}$, $P_e = \frac{(TP+FP)\times(TP+FN)+(FN+TN)\times(FP+TN)}{(TP+TN+FP+FN)^2}$, $Ka = \frac{P_0-P_e}{1-P_e}$
False Alarm (FA)	$FA = \frac{FP}{TN+FP}$
Missing Alarm (MA)	$MA = \frac{FN}{TP+FN}$
Total Error (TE)	$TE = \frac{TP+TN}{TP+TN+FP+FN}$
Precision (Pr)	$Pr = \frac{TP}{TP+FP}$

In the basic setting of each neural network maintains the same to guarantee comparative fairness, the training sample size of each neural network was consistent, the optimizer of the experiment was the Adam optimizer, and different initial values of the learning rate and its decay were selected by trial-and-error experiments. For quantitative comparison, seven widely used evaluation indicators were selected [2], as shown in Table II. These evaluation metrics are calculated based on the confusion matrix, which consists of four elements: true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs).

C. Comparative Results

According to the above experimental setting, two experiments were conducted to demonstrate the advantages of the proposed HFTN. The two experimental results are detailed below. The first experiment aims at testing the performance and robustness of the proposed HTFN, and the experimental results are presented in Figs. 5–11 and Tables III–VIII in terms of visual performance and quantitative comparisons, respectively. The experimental results indicate that the proposed HFTN-based neural network can achieve more advanced detection performance than the considered state-of-the-art methods. The visual comparisons in Figs. 5–11 show that the maps obtained by the proposed HFTN had fewer false and missed detection pixels than the other state-of-the-art methods. For example, as shown in Fig. 7(i), the false and missed detection pixels in the change detection map produced by our proposed HFTN network were 979 and 3124, respectively, which is the least number of false and missed detection pixels among the detection pixels obtained by the methods. The quantitative comparisons in Tables III–VIII support the observed visual performance. As shown in these quantitative comparisons, the proposed HTFN-based network achieved the best accuracies in terms of five out of seven evaluation indicators. For example, when we applied the proposed HTFN-based network on the Datasets A and B, the achieved improvements were 0.66% and 1.93% higher than that of the state-of-the-art methods in terms of OA for the Datasets A and B, respectively. In addition, the proposed HFTN-based network

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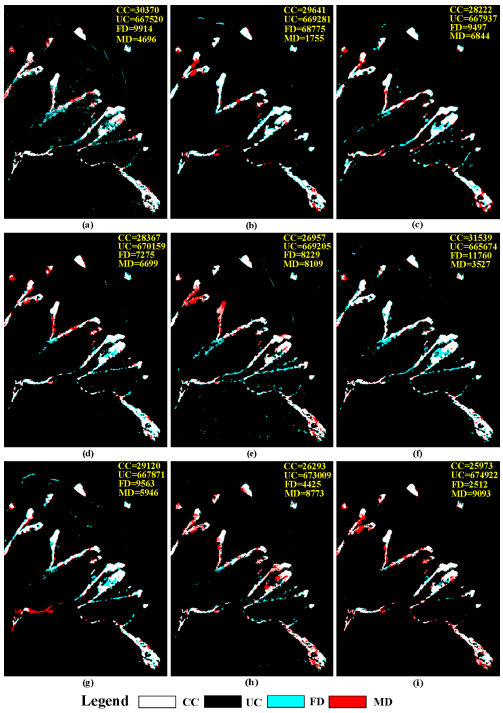


Fig. 6. Detection results obtained with the related methods for Dataset A. (a) SNUNET [33], (b) DSIFN [52], (c) BIT_CD [48], (d) MFCN [41], (e) PA-Former [47], (f) TFI-GR [67], (g) DSAMNet [68], (h) USSFC-Net [69], and (i) proposed (CC: correct change, UC: unchanged, FD: false detection, and MD: missed detection).

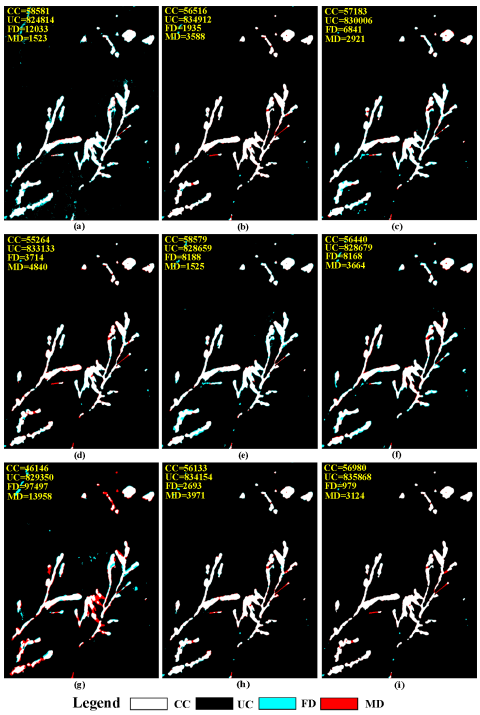


Fig. 7. Detection results obtained with the related methods for Dataset B. (a) SNUNET [33], (b) DSIFN [52], (c) BIT_CD [48], (d) MFCN [41], (e) PA-Former [47], (f) TFI-GR [67], (g) DSAMNet [68], (h) USSFC-Net [69], and (i) proposed (CC: correct change, UC: unchanged, FD: false detection, and MD: missed detection).

achieved the best precision (PR) results on all datasets. For example, the proposed method achieved precision improvements of 18.34 on Dataset A, 15.35 on Dataset B, and 17.10 on Dataset D compared to state-of-the-art approaches.

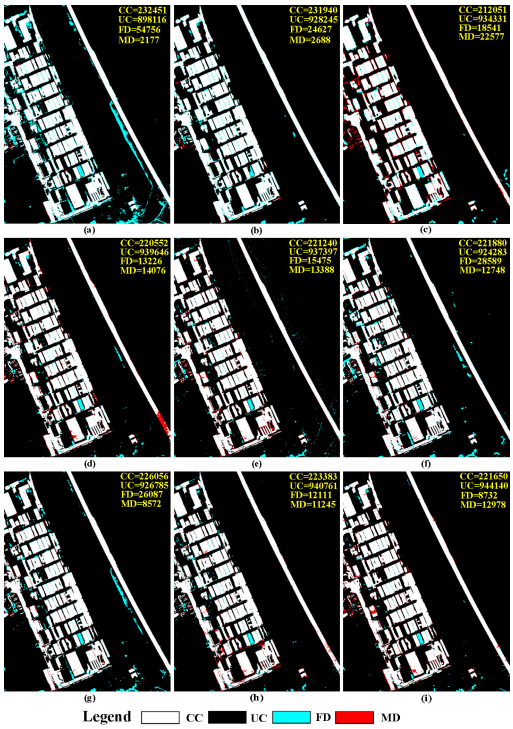


Fig. 8. Detection results obtained with the related methods for Dataset C. (a) SNUNET [33], (b) DSIFN [52], (c) BIT_CD [48], (d) MFCN [41], (e) PA-Former [47], (f) TFI-GR [67], (g) DSAMNet [68], (h) USSFC-Net [69], and (i) proposed (CC: correct change, UC: unchanged, FD: false detection, and MD: missed detection).

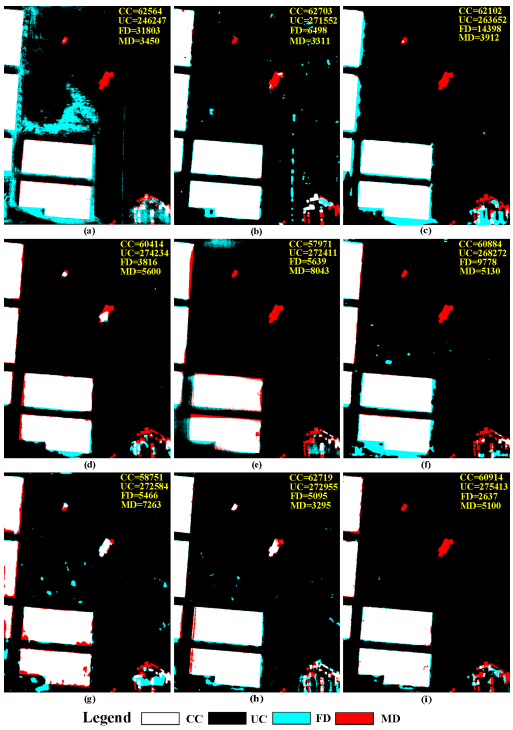


Fig. 9. Detection results obtained with the related methods for Dataset D. (a) SNUNET [33], (b) DSIFN [52], (c) BIT_CD [48], (d) MFCN [41], (e) PA-Former [47], (f) TFI-GR [67], (g) DSAMNet [68], (h) USSFC-Net [69], and (i) proposed (CC: correct change, UC: unchanged, FD: false detection, and MD: missed detection).

These significant improvements demonstrate the method's advantage in reducing false detections while maintaining high detection accuracy. Achieving optimal precision is

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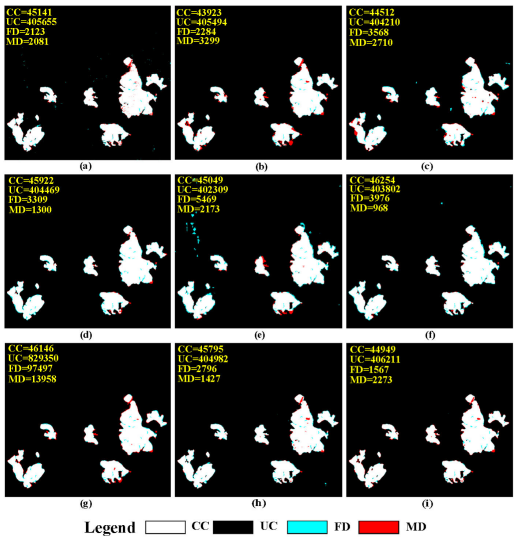


Fig. 10. Detection results obtained with the related methods for Dataset E. (a) SNUNET [33], (b) DSIFN [52], (c) BIT_CD [48], (d) MFCN [41], (e) PA-Former [47], (f) TFI-GR [67], (g) DSAMNet [68], (h) USSFC-Net [69], and (i) proposed (CC: correct change, UC: unchanged, FD: false detection, and MD: missed detection).

TABLE III

QUANTITIVELY COMPARISON FOR ALL THE METHODS IN TERMS OF THE SEVEN EVALUATION INDICATORS AND DATASET A

Method	OA	AA	Kappa	FA	MA	TE	Precision
SNUNET [33]	97.95	92.57	0.80	1.46	13.39	2.05	75.39
DSIFN [52]	98.09	91.66	0.80	1.20	15.47	1.91	78.43
BIT_CD [48]	97.71	89.54	0.76	1.40	19.52	2.29	74.82
MFCN [41]	98.04	89.91	0.79	1.07	19.10	1.96	79.59
PA-Former [47]	97.71	87.83	0.76	1.22	23.12	2.29	76.61
TFI-GR [67]	97.85	**94.10**	0.79	1.74	**10.06**	2.15	72.84
DSAMNet [68]	97.82	90.82	0.78	1.41	16.96	2.18	75.28
USSFC-Net [69]	98.15	87.16	0.79	0.65	25.02	1.85	85.59
Proposed	**98.37**	86.85	**0.81**	**0.37**	25.93	**1.63**	**91.18**

TABLE IV

QUANTITIVELY COMPARISON FOR ALL THE METHODS IN TERMS OF THE SEVEN EVALUATION INDICATORS AND DATASET B

Method	OA	AA	Kappa	FA	MA	TE	Precision
SNUNET [33]	98.49	98.01	0.89	1.44	**2.53**	1.51	82.97
DSIFN [52]	99.38	96.90	0.95	0.23	5.97	0.61	96.7
BIT_CD [48]	98.91	97.16	0.92	0.82	4.86	1.09	89.33
MFCN [41]	99.05	95.75	0.92	0.44	8.05	0.95	93.72
PA-Former [47]	98.92	**98.24**	0.92	0.98	2.54	1.08	87.75
TFI-GR [67]	98.68	96.46	0.90	0.98	6.10	1.32	87.37
DSAMNet [68]	97.61	87.94	0.80	0.89	23.22	2.39	86.04
USSFC-Net [69]	99.26	96.54	0.94	0.32	6.61	0.74	95.4
Proposed	**99.54**	97.34	**0.96**	**0.12**	5.20	**0.46**	**98.32**

very important in remote sensing change detection, because it ensures that the proposed method accurately identifies actual changes while minimizing false alarms (FAs). This is crucial for applications such as land cover monitoring, disaster management, and environmental changes detection, where false detections can lead to incorrect assessments and decision-making. In summary, the experimental results, including both visual comparisons and quantitative evaluations, demonstrate the effectiveness and accuracy of the proposed HFTN-based method.

TABLE V

QUANTITIVELY COMPARISON FOR ALL THE METHODS IN TERMS OF THE SEVEN EVALUATION INDICATORS AND DATASET C

Method	OA	AA	Kappa	FA	MA	TE	Precision
SNUNET [33]	95.21	96.66	0.86	5.75	**0.93**	4.79	80.94
DSIFN [52]	97.70	**98.13**	0.93	2.59	1.15	2.30	90.40
BIT_CD [48]	96.54	94.22	0.89	1.95	9.62	3.46	91.96
MFCN [41]	97.70	96.31	0.93	1.39	6.00	2.30	94.34
PA-Former [47]	97.57	96.33	0.92	1.62	5.71	2.43	93.4
TFI-GR [67]	97.08	96.80	0.91	2.74	3.65	2.92	89.65
DSAMNet [68]	98.03	96.97	0.94	1.27	4.79	1.97	94.86
USSFC-Net [69]	99.26	96.54	0.94	0.32	6.61	0.74	95.4
Proposed	**98.17**	96.78	**0.94**	**0.92**	5.53	**1.83**	**96.21**

TABLE VI

QUANTITIVELY COMPARISON FOR ALL THE METHODS IN TERMS OF THE SEVEN EVALUATION INDICATORS AND DATASET D

Method	OA	AA	Kappa	FA	MA	TE	Precision
SNUNET [33]	85.41	96.32	0.90	2.89	**4.46**	2.84	89.48
DSIFN [52]	97.15	96.32	0.91	2.34	5.02	2.85	90.61
BIT_CD [48]	94.68	94.45	0.84	5.18	5.93	5.32	81.18
MFCN [41]	97.26	95.07	0.91	1.37	8.48	2.74	94.06
PA-Former [47]	96.02	92.89	0.87	2.03	12.18	3.98	91.1
TFI-GR [67]	95.67	94.36	0.86	3.52	7.77	4.33	86.16
DSAMNet [68]	96.30	93.52	0.88	1.97	11.00	3.70	91.49
USSFC-Net [69]	97.56	**96.59**	0.92	1.82	4.99	2.44	92.49
Proposed	**97.81**	94.89	**0.93**	**0.37**	9.85	**2.19**	**98.28**

TABLE VII

QUANTITIVELY COMPARISON FOR ALL THE METHODS IN TERMS OF THE SEVEN EVALUATION INDICATORS AND DATASET E

Method	OA	AA	Kappa	FA	MA	TE	Precision
SNUNET [33]	99.08	97.54	**0.95**	0.52	4.41	0.92	95.51
DSIFN [52]	98.77	96.23	0.93	0.56	6.99	1.23	95.06
BIT_CD [48]	98.62	96.69	0.93	0.88	5.74	1.38	92.58
MFCN [41]	98.99	98.22	**0.95**	0.81	2.75	1.01	93.28
PA-Former [47]	98.32	97.03	0.91	1.34	4.60	1.68	89.17
TFI-GR [67]	98.91	**98.49**	0.94	0.98	**2.05**	1.09	92.08
DSAMNet [68]	98.83	96.70	0.94	0.62	5.99	1.17	94.64
USSFC-Net [69]	99.07	98.15	0.95	0.69	3.02	0.93	94.25
Proposed	**99.16**	97.40	**0.95**	**0.38**	4.81	**0.84**	**96.63**

The second experiment aims to investigate the performance and effectiveness of each submodule of the proposed HTFN network. To give a clear presentation, three proposed three submodules were denoted as FDM, PDM, and HFFM in an abbreviation way in the following ablation experiments. In addition, the backbone of the proposed approach is symbolized as "Base" in the ablation experiments.

Based on the above abbreviation, three pairs of datasets were selected for the experiments, and the ablation experiment's results are summarized and compared in Table IX. Table IX shows that each proposed submodule can effectively improve the accuracies of LCCD. For example, when we compared the performance of "Based" with that of "Based + FDM," it clearly demonstrated that the improvement can be achieved by FDM. Consequently, the results based on "Base + FDM + PDM" further indicate the effectiveness of the proposed PDM compared with those based on "Based + FDM," and "Based + FDM + FDM + HFFM" not only demonstrates the effectiveness of the HFFM but also indicates

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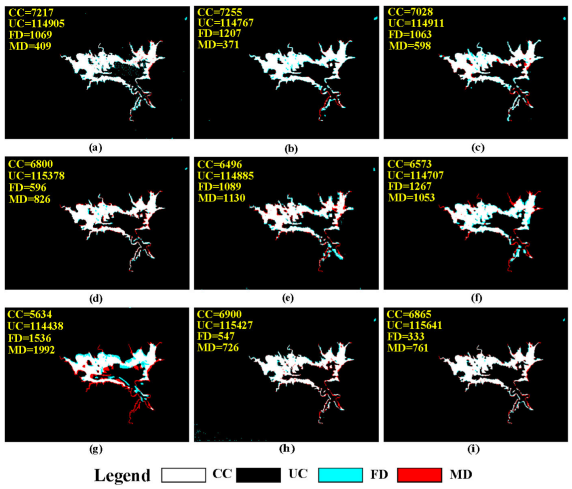


Fig. 11. Detection results obtained with the related methods for Dataset F. (a) SNUNET [33], (b) DSIFN [52], (c) BIT_CD [48], (d) MFCN [41], (e) PA-Former [47], (f) TFI-GR [67], (g) DSAMNet [68], (h) USSFC-Net [69], and (i) proposed (CC: correct change, UC: unchanged, FD: false detection, and MD: missed detection).

TABLE VIII

QUANTITIVELY COMPARISON FOR ALL THE METHODS IN TERMS OF THE SEVEN EVALUATION INDICATORS AND DATASET F

Method	OA	AA	Kappa	FA	MA	TE	Precision
SNUNET [33]	98.80	96.86	0.90	0.92	5.36	1.20	87.10
DSIFN [52]	98.72	**97.05**	0.90	1.04	**4.87**	1.28	85.74
BIT_CD [48]	98.66	95.62	0.89	0.92	7.84	1.34	86.86
MFCN [41]	98.85	94.33	0.90	0.51	10.83	1.15	91.94
PA-Former [47]	98.20	92.12	0.84	0.94	14.82	1.80	85.64
TFI-GR [67]	98.12	92.55	0.84	1.09	13.81	1.88	83.84
DSAMNet [68]	97.15	86.28	0.75	1.32	26.12	2.85	78.58
USSFC-Net [69]	98.97	95.00	0.91	0.47	9.52	1.03	92.65
Proposed	**99.16**	95.67	**0.93**	**0.35**	8.31	**0.84**	**94.50**

TABLE IX

ABLATION EXPERIMENTS OF THE PROPOSED HFTN

Dataset	Module	OA	AA	Kappa	FA	MA	TE	precision
Dataset-B	Base	99.20	96.14	0.93	0.33	7.39	0.80	95.26
	Base+FDM	99.27	96.11	0.94	0.24	7.55	0.73	96.57
	Base+FDM+PDM	99.45	96.94	**0.96**	0.16	5.96	0.55	97.70
	Base+FDM+PDM+HFFM	**99.54**	**97.43**	**0.96**	**0.12**	**5.20**	**0.46**	**98.32**
Dataset-E	Base	99.06	97.10	**0.95**	0.43	5.36	0.94	96.21
	Base+FDM	99.09	97.13	**0.95**	0.40	5.35	0.91	96.49
	Base+FDM+PDM	99.13	97.34	**0.95**	0.39	4.93	0.87	96.54
	Base+FDM+PDM+HFFM	**99.16**	**97.40**	**0.95**	**0.38**	**4.81**	**0.84**	**96.63**
Dataset-F	Base	99.04	94.31	0.91	0.30	11.08	0.96	95.15
	Base+FDM	99.07	**96.76**	0.92	0.32	10.15	0.93	94.80
	Base+FDM+PDM	99.11	94.96	0.92	**0.30**	9.78	0.89	95.17
	Base+FDM+PDM+HFFM	**99.16**	**95.67**	**0.93**	**0.35**	**8.31**	**0.84**	**94.50**

that the coupling of all the submodules can achieve better accuracies than using only one of the submodules.

D. Discussion

To expand the application of the proposed HTFN-based network in LCCD with RSIs, the relationship between the detection accuracies and the hyperparameters and the number of training samples is discussed below.

1) Hyperparameter λ_1 is adopted to adjust the influence of the mean square error and the cross-entropy loss functions. In each experiment, as shown in Fig. 12, for Datasets B and E, the precision decreased with the increase of λ_1. For example, the precision based on Dataset B decreased from 98.68% to 94.65% with an increment of $\lambda_1 = 0.2$ to $\lambda_1 = 0.9$.

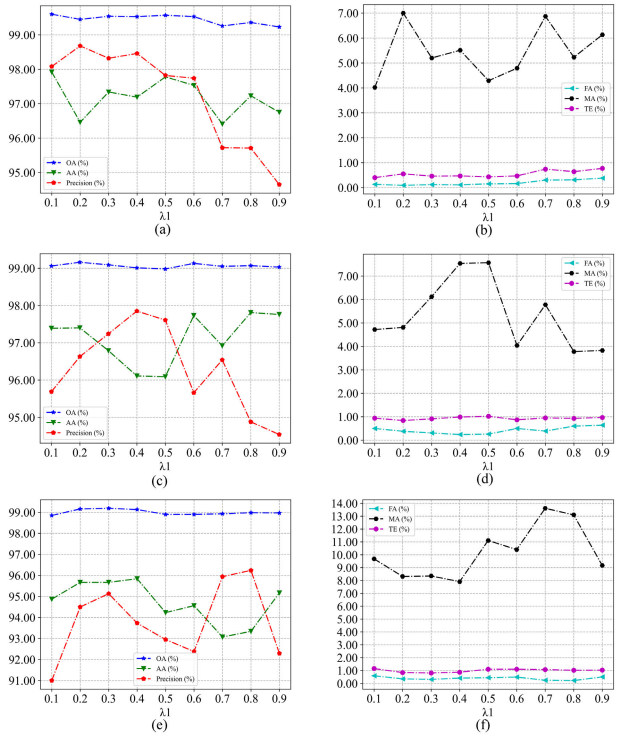


Fig. 12. Relationship between detection accuracies and the change of λ_1 for the proposed HTFN-based network. (a) and (b) Dataset B. (c) and (d) Dataset E. (e) and (f) Dataset F.

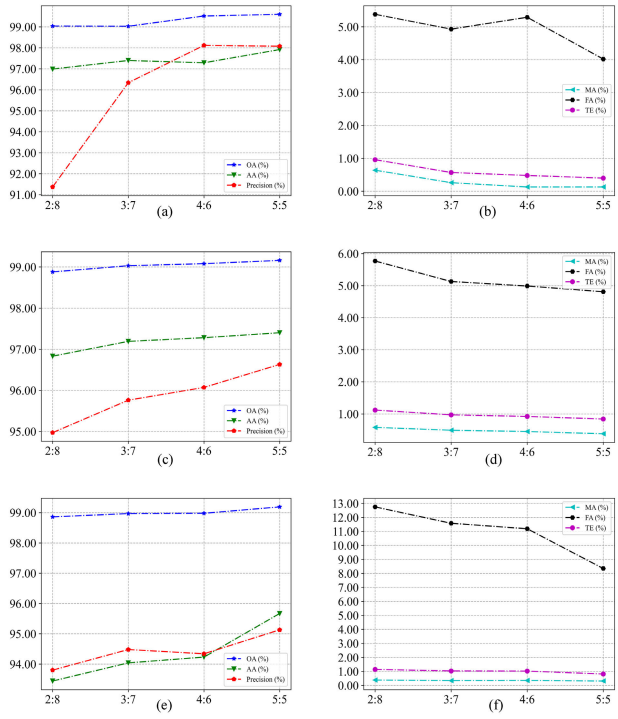


Fig. 13. Relationship between detection accuracies and the ratio between training samples and test samples for the proposed HTFN-based network. (a) and (b) Dataset B. (c) and (d) Dataset E. (e) and (f) Dataset F.

Meanwhile, the precision of Dataset F indicates that the precision varied in a disordered status with the increase of λ_1. Therefore, optical λ_1 for a specific dataset should be determined by trial-and-error experiments. The other evaluation indicators show that a similar trend for AA and MA can be obtained. In addition, the observed

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phenomenon shown in Fig. 12 reveals that the evaluation indicators, including OA, FA, and TE, are insensitive to the variation of λ_1 in the three datasets.

2) The relationship between the ratio of training and testing samples and the detection accuracies was investigated using the proposed HFTN network. As shown in Fig. 13, the ratio between the training and testing samples was set at 2:8, 3:7, 4:6, and 5:5. According to the experiments with the fixed ratio and numerous existing neural networks, the value of the evaluation indicators increased with the number of training samples for the proposed HFTN-based network, such as AA, precision, and FA. In addition, some evaluation indicators, such as OA, AA, MA, and TE, are insensitive to the change in the training and testing sample ratio, and the characteristics of the proposed HFTN may make it attractive in practical applications.

IV. Conclusion

In this article, an HFTN is proposed for LCCD with RSIs. In the proposed HFTN, three submodules are included, namely the FDM, the PDM, and the HFFM, to enhance the learning performance. Compared with eight state-of-the-art methods (i.e., SNUNET [33], DSIFN [52], BIT_CD [48], MFCN [41], PA-Former [47], TFI-GR [67], DSAMNet [68], and USSFC-Net [69]) and based on six pairs of real RSIs, the experimental results well indicated the feasibility and advantages of the proposed HFTN for LCCD. Moreover, the ablation experiments prove the performance of each proposed submodule. Therefore, the proposed HFTN not only has advantages in achieving better performance and higher detection accuracies but also the submodules of the HFTN may also be preferred in other neural networks for applications with RSIs.

Although the proposed HFTN has been tested with some real RSIs and compared with some state-of-the-art methods to verify its advantages, the proposed HFTN still has some limitations, which should be noted for practical applications. For instance, the HFTN is a supervised change detection neural network, and training samples are required for applications. Moreover, the optical composition of λ_1 and λ_2 for the proposed HFTN requires trial-and-error experiments on a dataset, but the process of which is time-consuming. In future research, we will improve the proposed HFTN to use an unsupervised neural network for LCCD with RSIs.

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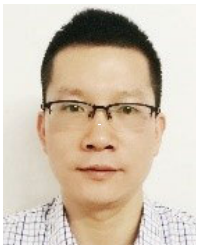
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